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Policy, Politics, & Nursing Practice
2025, Vol. 26(3) 219–226
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DOI: 10.1177/15271544251317506
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Lower Burnout Among Hospital Nurses in California Attributed to Better Nurse Staffing Ratios

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Abstract

Hospital nurses report that improving patient-to-nurse staffing ratios is a priority intervention to improve their well-being. In 2004, California became the first state to implement a hospital-wide policy mandating safe staffing ratios. This cross-sectional study determined whether nurses in California hospitals exposed to a staffing policy ("California advantage") experienced lower nurse burnout compared to those in hospitals not exposed, and whether part of the differences in burnout can be attributed to better hospital staffing. Nurse job outcomes (burnout [primary], job dissatisfaction, and intent to leave [secondary]), and nurse staffing variables were derived from the RN4CAST-US 2016 survey of 14,518 registered nurses in California, Florida, New Jersey, and Pennsylvania in 463 hospitals. Nurses in California had lower burnout, job dissatisfaction, and intentions to leave their employer, as compared to nurses in other states ($p < .001$). Mean patient-to-nurse staffing ratios were lower in California compared to non-California hospitals (3.8 vs. 4.7, $p < .001$). In bivariate logistic regression models, the California advantage was associated with lower odds of all nurse job outcomes (e.g., burnout $OR = 0.81$; 95% CI [0.74, 0.89]; $p < .001$). The California advantage, while smaller, remained statistically significant with the addition of nurse staffing in the model. Every patient added to a nurse's workload was associated with higher odds of nurse burnout (aOR = 1.12; 95% CI [1.06, 1.19]; $p < .001$), job dissatisfaction, and intent to leave. Better hospital nurse staffing partially mediated the California advantage on all job outcomes. California nurses experience better job outcomes, attributed in part to safer staffing due to a policy.

Keywords

workload, burnout, nurse, turnover

Introduction

One-third to half of hospital-based registered nurses report high burnout in the United States (Aiken, Sloane, et al., 2023; National Academies of Sciences, Engineering, and Medicine, 2019); and nurse burnout has been associated with poorer outcomes for patients, including higher odds of death, readmissions, and prolonged hospital stays (Aiken et al., 2002; Schlak et al., 2021). Evidence from quantitative and qualitative studies of nurses demonstrates that persistently high nurse workloads due to unsafe hospital patient-to-nurse staffing ratios contributes to the defining features of burnout: emotional exhaustion, depersonalization, and a

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sense of low personal accomplishment (McHugh et al., 2011; Muir et al., 2022; Schaufeli et al., 1996).

Nurse burnout is a key driver of nurse intentions to leave their employers as well as actual departures from employment (Aiken et al., 2002; Muir et al., 2024). While burnout can be driven by multiple occupational factors, nurses consistently rate poor staffing as a top reason for leaving healthcare and is the priority issue they recommend employers address to reduce their burnout (Aiken, Lasater, et al., 2023). A substantial literature, including landmark evidence from Aiken et al. (2002) demonstrates that every 1 patient added to a hospital nurse's workload is associated with 23% higher odds of burnout (Aiken, Lasater, et al., 2023; McHugh et al., 2011; Schlak et al., 2021).

California is the longest-standing U.S. state to implement hospital-wide patient-to-nurse staffing ratio legislation. Assembly Bill 394 mandates that hospitals adhere to safe nurse staffing ratios: for example, a ratio of five patients per nurse in medical-surgical units (California Legislative Information, 1999). Evidence from 2 years after the 2004 implementation of the legislation demonstrates that nurse staffing ratios were lower in California compared to non-California hospitals (Aiken et al., 2010). Although this study identified that patient outcomes would improve if non-California hospitals adhered to California staffing standards, it remains unknown whether nurse burnout is lower in California compared to non-California hospitals and whether the advantage of being exposed to a staffing policy in California contributes to lower burnout rates.

Addressing clinician burnout is a national priority identified by the U.S. Surgeon General and the National Academy of Medicine's Action Collaborative on Clinician Well-Being (National Academies of Sciences, Engineering, and Medicine, 2019; U.S. Surgeon General's Office, 2022). We hypothesized that nurse burnout would be better among California nurses who are exposed to a staffing policy ("California advantage") as compared to non-California nurses and that safer nurse staffing ratios in California would explain, at least in part, any differences in California nurses' job outcomes. If our hypotheses are confirmed, it would lend credence to an evidence-based policy intervention that has been successfully sustained for two decades and associated with reduced nurse burnout.

Methods

Design and Data

This was a cross-sectional study of hospital-based registered nurses that leveraged two data sources merged by a common hospital identifier: (a) the RN4CAST-US 2016 nurse survey which provided information on nurse job outcomes and staffing and (b) the 2016 American Hospital Association (AHA) Annual Hospital survey on hospital characteristics such as size and teaching status. The RN4CAST-US 2016 survey included a 30% random sample of registered nurses licensed to work in California, Florida, New Jersey, and Pennsylvania. A detailed overview of the RN4CAST-US 2016 survey has been reported elsewhere (Lasater et al., 2019). In short, nurses were mailed a survey to their home address and asked to report the name of their employer, unit (for hospital-based nurses), and role (e.g., direct care nurse, nurse manager). Nurses also provided information on their demographics, job outcomes (burnout, intent to leave their employer within the next year, job dissatisfaction), and aspects of nurse staffing in their employment setting.

The RN4CAST-US survey had a 26% initial response rate and a nonresponse survey yielded an 87% response rate (Lasater et al., 2019). There were no statistically significant differences in responses from the initial respondents and nonresponders on the majority of study measures. The 2016 AHA Annual Hospital Survey provided hospital characteristics (see "Covariates"; AHA, 2018). The study was approved by the University of Pennsylvania Institutional Review Board.

Sample

The analytic sample included 14,518 registered nurses employed in direct care positions working in any hospital unit. Extant literature demonstrates that a minimum of 10 nurse respondents per hospital produces reliable hospital-level estimates of nurse staffing (Lasater et al., 2019; McHugh & Ma, 2014). Therefore, hospitals were included in the study if they had at least 10 RN4CAST-US nurse respondents and participated in the 2016 AHA Annual Survey. The final analytic hospital sample was 463 non-federal, acute care hospitals in California, Florida, New Jersey, and Pennsylvania. Study hospitals had an average of 32.4 nurse respondents per hospital, ranging from 10 to 141.

Measures

Exposure. The exposure was a dichotomous variable indicating whether or not the nurse was employed in a hospital in California or a hospital in another state (i.e., Florida, New Jersey, or Pennsylvania). The state of employment was derived from the nurses' reports of their hospital name in the RN4CAST-US survey.

Nurse Job Outcomes. The primary study outcome was burnout and the secondary outcomes were job dissatisfaction and intent to leave from the RN4CAST-US study. Burnout was derived from the emotional exhaustion subscale of the Maslach Burnout Inventory (Aiken et al., 2002; Schlak et al., 2021). Nurses were classified as experiencing "high burnout" if their score was higher than the published top tertile for healthcare workers (≥27; Maslach et al., 2017).

Table 1. Demographic and Job Outcome Differences Among Hospital Nurses in California Relative to Hospital Nurses in Other States

	All nurses ($n = 14{,}518$)	CA nurses[a] ($n = 5{,}014$)	Non-CA nurses ($n = 9{,}504$)	p
Nurse demographics				
Age (years), *M* (*SD*)	46.5 (12)	46.7 (12)	46.4 (12)	0.13
Female, No. (%)	13,080 (90)	4,450 (89)	8,630 (91)	<.001
BSN or higher, No. (%)	8,184 (56)	3,044 (61)	5,140 (54)	<.001
Years of experience as a nurse, *M* (*SD*)	18.1 (13)	18.0 (12)	18.2 (13)	0.47
Nurse job outcomes				
High burnout, No. (%)	6,309 (43)	2,026 (40)	4,283 (45)	<.001
Job dissatisfaction, No. (%)	3,048 (21)	759 (15)	2,289 (24)	<.001
Intent to leave job, No. (%)	1,845 (13)	541 (11)	1,304 (14)	<.001

Note: BSN = Bachelor of Science in Nursing; No. = number; *SD* = standard deviation; Non-CA = FL, NJ, PA nurses; CA = California; FL = Florida; NJ = New Jersey; PA = Pennsylvania.
[a] Signifies exposure to nurse staffing policy mandate AB 394.

Job dissatisfaction was measured using a single-item question, "How satisfied are you with your primary job?" and dichotomized into satisfied (i.e., very satisfied/moderately satisfied) and dissatisfied (i.e., a little dissatisfied/very dissatisfied; Aiken, Lasater, et al., 2023). Intent to leave the employer was measured using a single yes or no question, "Do you plan to be with your current employer one year from now?"

Covariates

Nurse and Hospital Characteristics. Covariates included nurse characteristics: age, years of experience as a registered nurse, gender, and highest degree of educational attainment in nursing (dichotomized as whether or not a nurse had at least a Bachelor of Science in Nursing [BSN] or higher) from the RN4CAST-US survey.

Hospital characteristic data were from the 2016 AHA Annual Hospital Survey and included hospital size, teaching status, and technology capabilities. Hospital size was categorized into ≤100 beds (small), 101–250 beds (medium), and >250 beds (large). Teaching status was defined as nonteaching with no residents/fellows (0–4 medical trainees per bed), minor (0–4 medical trainees per bed), or major (≥4 medical trainees per bed). Technology capabilities were defined as hospitals with the capacity to perform open heart surgery and/or organ transplant.

Patient-to-Nurse Staffing Ratios. A hospital-level nurse staffing measure was generated from nurse responses in the RN4CAST-US survey. Nurse staffing was measured by direct care nurse reports of the total number of patients on their unit during the last shift, divided by the total number of registered nurses on the unit. A hospital-level patient-to-nurse staffing ratio measure was created by averaging the staffing ratio across all nurses in the same hospital.

Data Analysis

Differences in nurse demographics, job outcomes, and patient-to-nurse staffing ratios were compared for California and non-California nurses using chi-square tests for categorical variables and Student's *t*-tests for continuous variables. Hospital characteristics were similarly described across California and non-California states.

Multi-level logistic regression models were used to estimate the association between the California advantage (i.e., nurses' exposure to a staffing policy or not) and nurse job outcomes. A two-level model was specified with nurses nested within hospitals with a hospital-level random intercept. In our models, we assume that only hospitals in California were exposed to a staffing mandate aimed at providing safer patient-to-nurse staffing ratios (AB 394).

Bivariate models were first constructed (Model 1), followed by models accounting for hospital and nurse characteristics (Model 2). The final model (Model 3) accounted for hospital and nurse characteristics and additionally added the hospital-level patient-to-nurse staffing ratio variable (Model 3). The same sequential models were constructed for the secondary outcomes of nurse job dissatisfaction and intent to leave.

We additionally tested whether the effect of the California advantage on study outcomes was mediated by hospital nurse staffing. We performed the mediation analysis using the R *mediation* package (Imai et al., 2010; Tingley et al., 2014), which allows for the estimation of the mediation effect with multi-level models. Within the mediation model, we estimated three quantities: the total effect of the California advantage on outcomes; the direct effect; and the mediation effect: that is, the extent to which hospital nurse staffing accounts for the California advantage on outcomes. We acknowledge that, as Kline (2015) writes, "mediation analysis depends on many assumptions … Some of these assumptions are also untestable." While we cannot guarantee that all the assumptions needed for this analysis are met; the

adopted study design (including the timing of the passing of the California staffing mandate in relation to the timing of when the outcomes and staffing were measured) and analytical methodology provide a high degree of assurance about the credibility and robustness of the reported results.

Results

Descriptives

Among the total sample of 14,518 nurses, 5,014 were employed in a California hospital (e.g., exposed to a staffing policy mandate), while 9,504 were employed in a non-California hospital (Table 1). California and non-California nurses were similar with respect to age ($M = 46.7$ years vs. 46.4, $p = .13$) and years of experience as a nurse ($M = 18.0$ years vs. 18.2 years, $p = .47$). Nurses employed in California hospitals were less likely to be female compared to California nurses (89% vs. 91%, $p \leq .001$), though the difference is not substantively significant; and were more likely to have a BSN degree or higher (61% vs. 54%, $p \leq <.001$). California nurses when compared to non-California nurses reported lower rates of high burnout (40% vs. 45%, $p < .001$), job dissatisfaction (15% vs. 24%, $p < .001$), and intent to leave (11% vs. 14%, $p < .001$).

Of the 463 study hospitals, 183 hospitals were in California (exposed to a staffing policy mandate) and the remaining 280 were distributed across the three other states (129 in Florida, 49 in New Jersey, and 102 in Pennsylvania; Table 2). Patient-to-nurse staffing ratios were lower in California compared to non-California hospitals; such that on average, nurses in California cared for nearly one fewer patient compared to nurses in other states ($M = 3.8$ vs. 4.7, $p < .001$). The distribution of nurse staffing ratios across hospitals in California and other states is displayed in Figure 1. There were no significant differences between California and non-California hospitals with respect to hospital size, teaching status, or technology capabilities.

The logistic regression results are provided in Table 3. The unadjusted models (Model 1) demonstrate that the California advantage was associated with lower odds of nurse burnout, job dissatisfaction, and intent to leave. For example, nurses' employment in a California hospital was associated with 19% lower odds of burnout as compared to non-California nurses. The lower odds of all nurse job outcomes remained the same after accounting for differences in nurse and hospital characteristics (Model 2). In the fully adjusted model (Model 3), hospital nurse staffing was associated with all job outcomes and slightly attenuated the effect of the California advantage on the outcomes. For example, each additional patient per nurse was associated with a 12% increase in the odds of nurse burnout. Accounting for hospital-level staffing differences partially reduced the California advantage on lower nurse burnout, job dissatisfaction, and intent to leave. Full model regression results are provided in Supplementary Appendix Table 1.

Table 2. Characteristics of the Hospital Sample ($n = 463$ Hospitals)

Hospital characteristic, *n* (%)	All hospitals ($n = 463$)	CA hospitals ($n = 183$)	Non-CA hospitals ($n = 280$)	*p*
Bed size				.65
Small (<100)	17 (4)	5 (3)	12 (4)	
Medium (101–250)	187 (40)	73 (40)	114 (41)	
Large (>250)	259 (56)	105 (57)	154 (55)	
Technology status				.07
High technology	270 (58)	116 (63)	154 (55)	
Non-high technology	193 (42)	67 (37)	126 (45)	
Teaching status				.12
Major	46 (10)	13 (7)	33 (12)	
Minor	224 (48)	85 (47)	139 (49)	
None	193 (42)	85 (46)	108 (39)	
Mean patient-to-nurse staffing ratio	4.4 (0.9)	3.8 (0.8)	4.7 (0.9)	<.001

Note: CA = California. Mean patient-to-nurse staffing ratios assess staffing on all hospital units.

Mediation Analysis

The multi-level model findings are contextualized further by a supplemental analysis (see Supplementary Appendix Table 2) demonstrating that hospital nurse staffing partially mediated the California advantage on all three study outcomes. Specifically, we found that hospital nurse staffing accounts for about 50% of the total effect of the California policy advantage on nurse burnout, and 19% and 37% for job dissatisfaction and intent to leave, respectively.

Discussion

Our study extends existing evidence that hospital nurses in California are less likely to report high burnout, job dissatisfaction, and intentions to leave their employer compared to nurses in other states. That California nurses, on average, care for one fewer patient than non-California nurses explain about 50% of the California advantage for lower burnout, and a bit less for lower job dissatisfaction and intent to leave. The mediation analysis shows that safer nurse staffing ratios significantly reduce, but do not eliminate the California advantage effect on outcomes.

Our findings are informative to national organizations, healthcare leaders, and legislators actively seeking evidence-based solutions to reduce nurse burnout in the United States

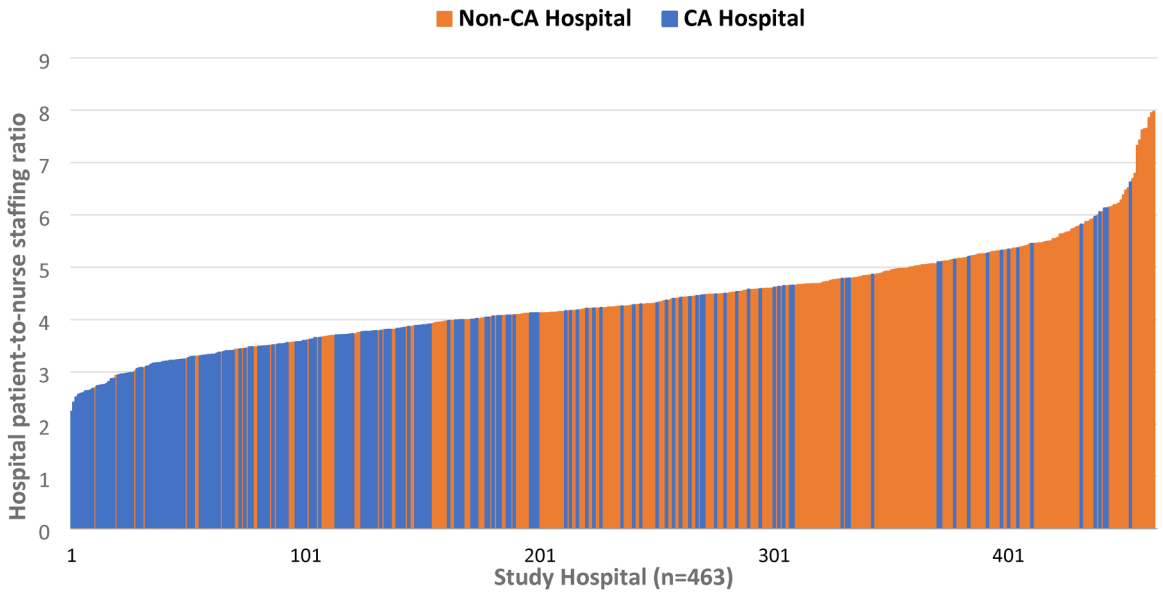

Figure 1. Mean hospital patient-to-nurse staffing ratio in California and non-California hospitals.
Note. Darker shade (blue) indicates California hospitals and lighter shade (orange) indicates non-California hospitals.

Table 3. Unadjusted and Adjusted Models Estimating the Association Between the California Advantage and Nurse Job Outcomes.

Nurse outcomes	Model 1	Model 2 (controlling for nurse and hospital characteristics)	Model 3 (controlling for nurse and hospital characteristics + hospital nurse staffing)
	OR [95% CI]	*OR* [95% CI]	*OR* [95% CI]
High burnout			
CA advantage	0.81 [0.74, 0.89]***	0.81 [0.74, 0.89]***	0.90 [0.81, 1.00]*
Staffing	–	–	1.12 [1.06, 1.19]***
Job dissatisfaction			
CA advantage	0.53 [0.47, 0.60]***	0.54 [0.48, 0.61]***	0.60 [0.53, 0.69]***
Staffing	–	–	1.13 [1.05, 1.22]**
Intent to leave			
CA advantage	0.73 [0.64, 0.84]***	0.72 [0.62, 0.82]***	0.81 [0.69, 0.94]*
Staffing	–	–	1.14 [1.05, 1.24]**

Note: The California advantage signifies nurses' exposure to a hospital staffing policy through employment in a California hospital or not. Model 1 is unadjusted. Model 2 is adjusted for individual nurse characteristics (age, gender, education, years of experience) and AHA hospital characteristics (bed size, teaching status, technology status). Model 3 is adjusted for individual nurse characteristics, AHA hospital characteristics, and hospital-level nurse staffing ratios. Policy exposure refers to nurses in CA. CI = confidence interval; CA = California; AHA = American Hospital Association.
*$p < .05$. **$p < .01$. ***$p < .001$.

(National Academies of Sciences, Engineering, and Medicine, 2019; U.S. Surgeon General's Office, 2022). We evaluated nurse job outcomes in California, where patient-to-nurse staffing ratios are significantly lower as a result of Assembly Bill 394 enacted in 2004. Our finding that lower nurse burnout is associated with more favorable staffing ratios is supported by a substantial literature (Aiken et al., 2002; Helfrich et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019; Shah et al., 2021). The results from our study demonstrate that 20 years since the passage of California staffing policy, nurses in California hospitals have a sustained advantage over hospital nurses in other states with respect to better job outcomes (Lasater et al., 2021).

Nurses in California exposed to a nurse staffing mandate reported lower burnout, job dissatisfaction, and intent to leave compared to nurses in other states in this study. Nurse staffing was also significantly associated with higher odds of nurse burnout, job dissatisfaction, and intent to leave. Although nurse staffing attenuated the California advantage more on burnout than on job dissatisfaction and intent to leave; the California advantage remained significantly associated with all the outcomes. These findings suggest that there are other factors beyond nurse staffing that may explain the better rates of nurses' burnout, job dissatisfaction, and intent to leave among California nurses (Muir et al., 2024).

There are work environment factors that are unmeasured in the present study that may be related to the California staffing policy and contribute to better job outcomes among California nurses. For example, better staffing due to the California staffing policy may provide nurses with greater command over their work leading to more job satisfaction. Improved relationships between front-line nurses and management due to safer workloads may reduce a nurses' intent to leave the job (Bakker et al., 2023; Greenberger et al., 1989). Burnout was explained more by staffing as compared to other job outcomes, potentially because the California staffing policy is a protective buffer against emotional exhaustion, depersonalization, and a sense of low accomplishment. With safer workloads, nurses may be able to take more breaks on the job, experience more predictable scheduling, and provide safer care that mitigates against cynicism and a detachment from work (Bakker et al., 2023; Greenberger et al., 1989). Future explorations of specific work environment features that may explain the California advantage on job outcomes beyond staffing are needed.

Using contemporary data almost 12 years after California enacted legislation mandating hospital-wide minimum nurse staffing requirements, the findings demonstrate that hospital nurses in California, on average, have an advantage over nurses in other states with respect to better patient-to-nurse staffing ratios as well as better nurse job outcomes, like lower burnout (Aiken et al., 2010).

Minimum nurse staffing requirements are an evidence-based policy that address what nurses report to be their priority intervention to reduce burnout and improve their well-being (Aiken, Lasater, et al., 2023). States are increasingly considering minimum nurse staffing requirements given the evidence establishing a relationship between poor hospital nurse staffing and patient (e.g., higher odds of death) and nurse outcomes (e.g., burnout; Cimiotti et al., 2022). In 2024, Oregon became the second state to enact nurse staffing legislation mandating minimum nurse staffing requirements in acute care hospitals (Oregon Association of Hospitals and Health Systems, n.d.). Other states, including New York, Illinois, Maine, Massachusetts, and Pennsylvania, have introduced nurse staffing legislation as national organizations such as the American Nurses Association (American Nurses Association, 2023) and leaders of large U.S. health care systems express support for these evidence-based policies (Mahoney & Aiken, 2023). Minimum nurse staffing requirements have also been successfully enacted internationally in Queensland, Australia where they have been shown to be associated with lower mortality rates, readmission, and length of stay (McHugh et al., 2020).

The findings from this study demonstrate the sustained California advantage of safer nurse staffing ratios and lower nurse burnout, using data collected almost 10 years after prior research reported similar findings (Aiken et al., 2010).

Limitations

This report uses an observational study design, thus causal relationships cannot be inferred from the findings. Specifically, there might be unmeasured confounders in the relationship between the California advantage, nurse staffing ratios, and job outcomes that could explain some of the differences in outcome between California and non-California nurses. Our findings extend existing evidence (Aiken et al., 2010) by using a multi-level modeling design supplemented by a mediation analysis to further understand the extent to which better job outcomes among California nurses is attributed to safer hospital staffing. The study data were collected before the COVID-19 pandemic; however, contemporary research demonstrates that nurse burnout, job dissatisfaction, and intent to leave reports among nurses did not substantially change in the periods before and during the pandemic—and in fact, nurses working in hospitals that were better staffed prior to the pandemic reported better job outcomes during the pandemic (Aiken, Lasater, et al., 2023). Furthermore, more recent data collected during the pandemic (if available) would not be reflective of the staffing mandate policy, as California waved the legislation temporarily due to COVID-19.

Conclusions

Nurses in California report lower burnout, job dissatisfaction, and intent to leave compared to non-California nurses. About 50% of the differences in California nurses' burnout was attributed to better patient-to-nurse staffing ratios in California hospitals. Nurse staffing legislation is an evidence-based policy that, if leveraged, has the potential to reduce nurse burnout and improve retention of nurses in U.S. hospitals.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was funded by grants to University of Pennsylvania's Center for Health Outcomes and Policy Research from the National Institutes of Nursing Research (R01NR014855 awarded to McHugh; T32NR007104 for support of Muir, Sliwinski, Pogue, and Petto).

Ethical Approval

This study was approved by the University of Pennsylvania Institutional Review Board.

Supplemental Material

Supplemental material for this article is available online.

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